

Assignment – Week 4 Introduction to Machine Learning-IIT KGP

Topics: Probability Basics, Bayesian Learning, Naive Bayes, Bayesian Networks

TYPE OF QUESTION: MCQ

Number of Questions: 10 Total

Marks: $10 \times 2 = 20$

Question 1:

A box contains 4 red balls and 6 blue balls. A sensor correctly identifies a red ball with probability 0.8 and incorrectly identifies a blue ball as red with probability 0.1. If the sensor reports red, what is the probability the ball is actually red?

- A. 0.65
- B. 0.84
- C. 0.58
- D. 0.72

Correct Answer: B Detailed

Solution:

Let: R = Ball is actually red, B = Ball is actually blue, S = Sensor reports red Given:

$P(R) = 4/10 = 0.4$, $P(B) = 6/10 = 0.6$, $P(S|R) = 0.8$, $P(S|B) = 0.1$

Using Bayes' Theorem:

$$P(R|S) = [P(S|R) \times P(R)] / [P(S|R) \times P(R) + P(S|B) \times P(B)]$$

$$= (0.8 \times 0.4) / [(0.8 \times 0.4) + (0.1 \times 0.6)]$$

$$= 0.32 / 0.38$$

$$= 0.842$$

$$\approx 0.84$$

Therefore, the probability that the ball is actually red given that the sensor reports red is 0.84

Question 2:

Two fair dice are rolled. Given that the sum is 8, what is the probability that the first die shows 3?

- A. $1/5$
- B. $1/4$
- C. $1/6$
- D. $1/3$

Correct Answer: A

Detailed Solution: Possible outcomes are (2,6),(3,5),(4,4),(5,3),(6,2). Only one has the first die = 3.

Question 3:

A Naive Bayes classifier is used to classify emails as Spam (S) or Not Spam (N). The prior probabilities of the two classes are: $P(S) = 0.4$, $P(N) = 0.6$. For an incoming email, two features are observed:

- F_1 : The email contains the word "Offer"
- F_2 : The email contains the word "Free"

The following conditional probabilities are known:

$$P(F_1|S) = 0.8$$

$$P(F_2|S) = 0.7$$

$$P(F_1|N) = 0.3$$

$$P(F_2|N) = 0.2$$

Assuming that the Naive Bayes independence assumption holds and that the email contains both features F_1 and F_2 , how will the email be classified?

- A. Spam (S)
- B. Not Spam (N)
- C. Both classes are equally likely
- D. Cannot be determined from the given information

Correct Answer: A. Spam (S)

Detailed Solution:

Using the Naive Bayes classifier, we compute a score for each class. For Spam (S):

$$\begin{aligned}\text{Score}(S) &= P(S) \times P(F_1|S) \times P(F_2|S) \\ &= 0.4 \times 0.8 \times 0.7 \\ &= 0.224\end{aligned}$$

For Not Spam (N):

$$\begin{aligned}\text{Score}(N) &= P(N) \times P(F_1|N) \times P(F_2|N) \\ &= 0.6 \times 0.3 \times 0.2 \\ &= 0.036\end{aligned}$$

Comparing the two scores:

$$\text{Score}(S) = 0.224$$

$$\text{Score}(N) = 0.036$$

Since $\text{Score}(S) > \text{Score}(N)$, the classifier predicts the email as Spam.

Therefore, the correct answer is: Spam (S).

Question 4:

In a Bayesian Network, what does a directed edge from node A to node B indicate?

- A. A and B are always independent.
- B. A directly influences or provides probabilistic information about B.
- C. B causes A.
- D. A and B must have the same probability distribution.

Correct Answer: B. A directly influences or provides probabilistic information about B.

Detailed Solution: A directly influences or provides probabilistic information about B

Question 5:

A disease affects 5% of the population.

A diagnostic test is used to detect the disease. The test has the following characteristics:

$$P(+|D) = 0.95$$

$$P(+|\neg D) = 0.10$$

where:

- D denotes that a person has the disease.
- $\neg D$ denotes that a person does not have the disease.
- + denotes a positive test result.

If a randomly selected person tests positive, what is the probability that the person actually has the disease, i.e., $P(D|+)$?

- A. 0.33
- B. 0.50
- C. 0.67
- D. 0.90

Correct Answer: A. 0.33

Detailed Solution:

We are given:

$$P(D) = 0.05, P(\neg D) = 1 - 0.05 = 0.95, P(+|D) = 0.95, P(+|\neg D) = 0.10$$

We need to compute: $P(D|+)$

Using Bayes' Theorem:

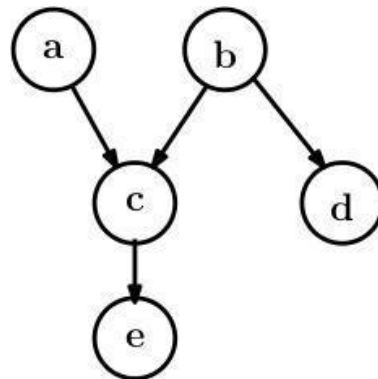
$$P(D|+) = [P(+|D) \times P(D)] / P(+)$$

$$P(+)=P(+|D)P(D)+P(+|\neg D)P(\neg D)=(0.95\times 0.05)+(0.10\times 0.95)=0.0475+0.095=0.1425$$

$$\text{Now apply Bayes' Theorem: } P(D|+)=(0.95\times 0.05)/0.1425=0.0475/0.1425=0.3333\approx 0.33$$

Question 6:

Consider the following graphical model, mark which of the following pair of random variables are independent given no evidence?



- A. a,b
- B. c,d
- C. e,d
- D. c,e

Correct Answer : A. a,b

Detailed Solution : Nodes a and b don't have any predecessor nodes. As they don't have any common parent nodes, a and b are independent.

Question 7:

Which statement is true regarding Naive Bayes?

- A. Features are independent of the class label.
- B. Features are conditionally independent given the class label.
- C. Classes are independent of features.
- D. Features are always uncorrelated.

Correct Answer: B

Detailed Solution: Naive Bayes assumes conditional independence given the class.

Question 8:

Consider a Bayesian network: Rain→WetRoad

Given $P(\text{Rain})=0.4$, $P(\text{WetRoad}|\text{Rain})=0.9$ and $P(\text{WetRoad}|\neg\text{Rain})=0.2$. Find $P(\text{WetRoad})$.

- A. 0.48
- B. 0.62
- C. 0.72
- D. 0.90

Correct Answer: A

Detailed Solution: $0.9 \times 0.4 + 0.2 \times 0.6 = 0.48$.

Question 9:

A fair coin is tossed three times. What is the probability of obtaining exactly two heads?

- A. $1/8$
- B. $3/8$
- C. $1/2$
- D. $5/8$

Correct Answer: B

Detailed Solution:

Number of outcomes with exactly two heads: ${}^3C_2 = 3$

Total no. of outcomes = $2^3 = 8$

So, probability of obtaining exactly two heads = $3/8$

Question 10:

What is the naive assumption in a Naive Bayes Classifier?

- A. All the classes are independent of each other
- B. All the features of a class are independent of each other
- C. The most probable feature for a class is the most important feature to be considered for classification
- D. All the features of a class are conditionally dependent on each other.

Correct Answer: B. All the features of a class are independent of each other

Detailed Solution: Naive Bayes Assumption is that all the features of a class are independent of each other.